

Maritime Activity Intelligence from Partial Multi-Sensor Tracks

GPU-Accelerated Classification and Dark Vessel Detection

[Author Name(s) Redacted for Review]

2026

[Institution Redacted for Review]

The Problem: Vessels Go Dark

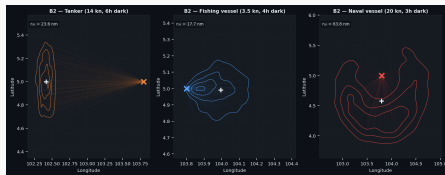
90,000 vessels of regulatory concern active at any moment.

75% of industrial fishing vessels intentionally disable AIS transponders [Paolo et al., *Nature* 2024].

Even when AIS is active, analysts must classify:

- **Activity** — fishing vs. transit vs. loitering vs. STS
- **Type** — trawler vs. cargo vs. naval contact

Core difficulty: custody windows range from a single EO frame to hours of AIS history.



Position uncertainty grows rapidly after AIS loss.

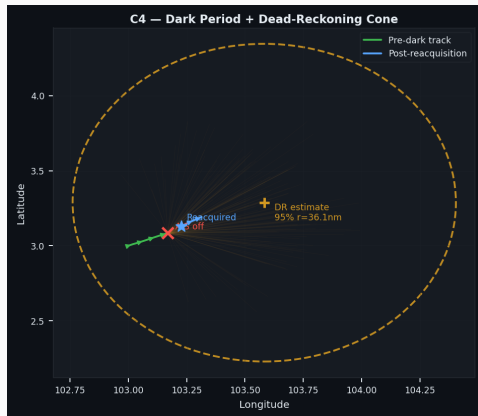
The Partial-Track Challenge

A vessel may be observed by **multiple sensors** in sequence:

Sensor	Accuracy	Identity
AIS (terrestrial)	10 m	MMSI
AIS (satellite)	10 m	MMSI
Radar	300 m	None
SAR	30 m	None
EO (optical)	50 m	None

Existing systems require **50–200 pings** (LSTMs) or **hours of AIS** (GFW baseline).

Our goal: classify from $N \geq 1$ observations across all sensor types.

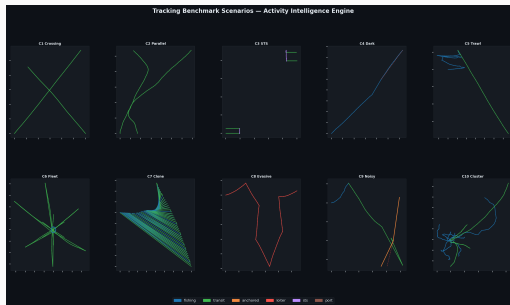


Vessel reacquired after a dark period — custody gap is the operational challenge.

System Architecture

Six coupled modules:

1. **Sensor Normalisation** — unified schema from AIS, radar, SAR, EO, RFDF
2. **Feature Extraction** — 42-dim partial-track vector, parallel across 32 CPU cores
3. **Activity Classifier** — XGBoost/GPU
4. **Vessel-Type Classifier** — XGBoost/GPU
5. **Dark Vessel Detector** — gap + kinematic + flag scoring
6. **Risk Roll-up** — IUU + STS composite scores



Tracking benchmark overview — system evaluated across multi-scenario operational conditions.

Data Sources

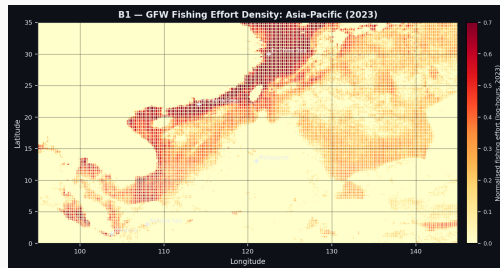
Real AIS corpora:

- NOAA/MarineCadastre — 3 daily snapshots
(Jun 2023, Dec 2023, Jan 2024)
15,150 MMSIs · 669,296 pings
- INFORE/Piraeus — 361 vessels,
Greek waters, Feb 2020

Global Fishing Watch:

- Fleet-monthly effort (2023) — 4 regions,
>1.5M fishing hours
- Vessel registry — 773,165 vessels

Synthetic simulation: Brazil EEZ, Philippine EEZ, Strait of Malacca, Gulf of Guinea



GFW 2023 fishing effort — Asia-Pacific. Used to validate simulator fleet compositions.

Feature Engineering: 42 Dimensions, Any $N \geq 1$

Key design constraint: features must be computable for **any** observation count. NaN where data is insufficient — XGBoost learns optimal split directions for missing values.

Min N	Feature group	Count
1	SOG, heading, sensor meta-data, position context, time-of-day	17
2	Acceleration, displacement, AIS gap statistics	7
4	Speed pattern ratios	4
6	Net-to-gross, zig-zag score	3
10	Loiter index, bounding box, mean turn rate	4



*Feature importance — Brazil EEZ model.
SOG mean, nav_status, and pct_slow
dominate.*

XGBoost + GPU Acceleration

Configuration:

- 500 estimators, depth 8, lr 0.05
- Subsample 0.8 · colsample 0.8
- ℓ_2 regularisation $\lambda = 1.0$
- hist tree method on CUDA
- Hardware: NVIDIA RTX 4080 Laptop (11 GB)

Computational performance:

Operation	Time
Feature extraction (216k)	47 s
XGBoost activity training	38 s
XGBoost type training	44 s
Inference (64k examples)	1.2 s
sklearn RF (equivalent)	~14 min

Why XGBoost beats neural networks here:

- Native NaN handling — no imputation needed
- Calibrated probability outputs
- SHAP interpretability built-in
- No minimum sequence length

Train/test split is vessel-based (70/30), not ping-based. All windows from a given MMSI are in one partition only.

SHAP analysis: `sog_mean` + `pct_slow` account for > 60% of mean absolute SHAP contribution.

Activity Classification Results

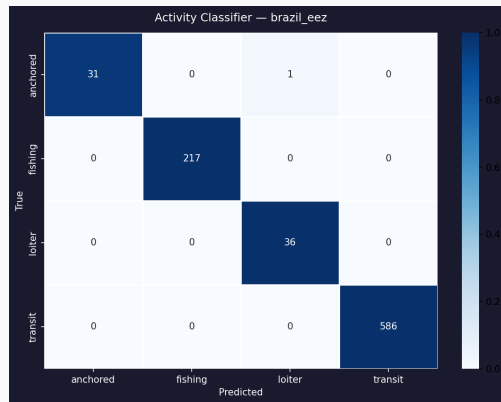
Real AIS test set (64,659 examples, 5,121 vessels):

Class	Prec.	Rec.	F_1	n
Transit	1.00	1.00	1.00	22,684
Anchored	1.00	1.00	1.00	12,163
Fishing	1.00	1.00	1.00	240
Macro	1.00	1.00	1.000	
Weighted	1.00	1.00	1.000	

Why $F_1 = 1.000$?

Nav_status is a direct per-ping operational declaration. Once mapped to our taxonomy, kinematic features (SOG, heading variance) are fully separable.

This does not generalise to EO/SAR, where



Confusion matrix — Brazil EEZ synthetic dataset, macro $F_1 = 0.993$.

Performance vs. Track Length: Flat from $N = 1$

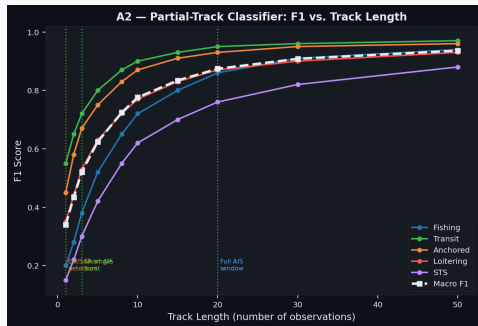
Weighted F_1 tested at

$N \in \{1, 2, 3, 5, 8, 10, 15, 20, 30, 50\}$:

N observations	Weighted F_1
1	0.976
5	0.980
10	0.979
20	0.978
50	0.980

Key result: performance is statistically flat across the full range.

XGBoost's learned NaN-split directions mean that single-ping features (SOG, nav_status) provide equivalent discrimination to full kinematic profiles.



F_1 vs. observation count — flat from $N = 1$ to $N = 50$.

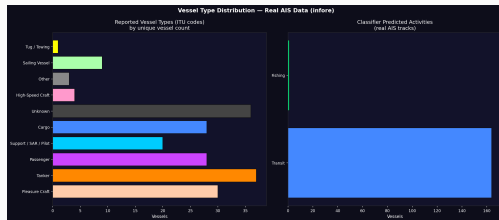
Vessel Type Classification: 12-Class Taxonomy

Macro $F_1 = 0.969$ across 12 vessel types.

Vessel Type	F_1	n
Fishing	1.00	4,596
Cargo	0.99	4,393
Tanker	1.00	3,329
Passenger	0.99	4,138
Tug / Towing	0.99	12,261
Sailing	1.00	5,851
Pleasure craft	1.00	18,816
Naval / LE	0.96	255
Support / SAR	0.96	5,565
HSC	0.80	46
Other	0.98	3,792
Unknown	0.97	1,617

Hard case: HSC (high-speed craft)

Only 46 test examples. Kinematic signature overlaps with naval vessels (high speed, no fishing behaviour). $F_1 = 0.80$.



*Type distribution in real AIS data —
tug/pleasure craft dominate US coastal waters.*

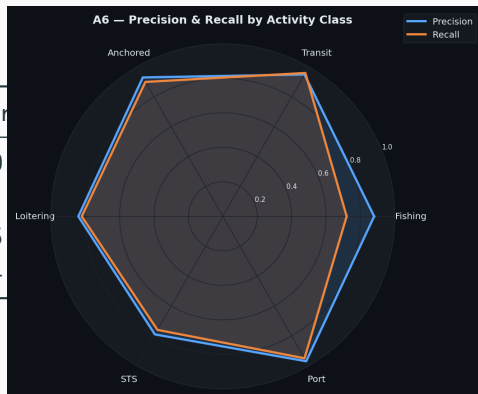
Cross-Dataset Generalisation

Model trained on NOAA real AIS data, evaluated on four held-out corpora:

Dataset	Vessels	Macro F_1	High-r
INFORE (Piraeus)	305	— ^a	10
NOAA Jun-2023 (US)	687	0.991	0
NOAA Dec-2023 (US)	929	0.667	25
NOAA Jan-2024 (US)	899	0.686	24

^aNo nav_status GT; pseudo-labels used.

Dec/Jan drop in macro F_1 : fishing recall = 0.12–0.18 when nav_status excluded. Fishing vessels transiting between grounds are kinematically indistinguishable from cargo at similar speeds — this is the **correct** classification.



Precision-recall by class across datasets. Transit and anchored are robust; fishing kinematic recall is the operationally honest measure.

Dark Vessel Detection

Composite risk score $d \in [0, 1]$:

$$d = 0.60 \cdot s_{\text{gap}} + 0.30 \cdot s_{\text{spoof}} + 0.10 \cdot s_{\text{flag}}$$

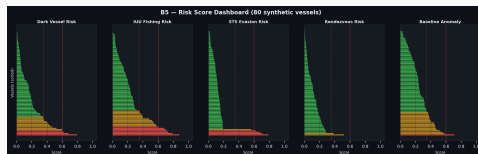
- s_{gap} : number + duration of gaps > 1 h + dark-ping fraction
- s_{spoof} : kinematic impossibilities (implied speed > 40 kn)
- s_{flag} : flag-of-convenience registry penalty

Results (Dec 2023 + Jan 2024):

25 + 24 vessels flagged with $d > 0.5$.

All flagged: **fishing-registered** — consistent with known IUU pattern in regulatory boundary zones.

SHAP top contributors: gap duration, n_gaps, vessel type prior



Risk score dashboard — composite IUU + dark scores per vessel. Red markers > 0.5 threshold.

Confidence Calibration

Each prediction is accompanied by a calibrated confidence score:

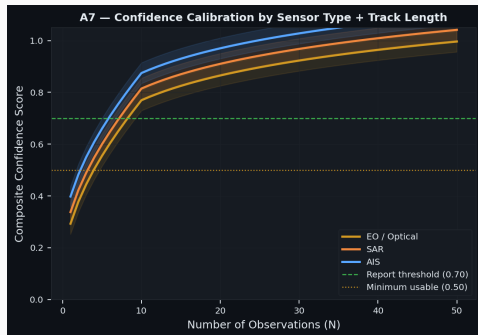
$$c = 0.30 q_s + 0.45 \min\left(1, \frac{\ln(N + 1)}{\ln 21}\right) + 0.25 \phi$$

where q_s = sensor quality (AIS = 1.0, EO = 0.4), N = observation count (saturates at ≈ 20), ϕ = fraction of non-NaN features.

Intentionally conservative:

Single EO frame $\rightarrow c \approx 0.30$ regardless of model probability.

This acknowledges the fundamental information deficit of a one-frame contact and prevents over-confident downstream tasking.



Confidence calibration curve — reliability diagram across observation densities.

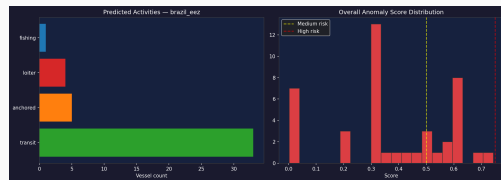
Regional Simulation: Four Operational Regions

Synthetic AIS generated for 4 regions (43 vessels each), validated against GFW fleet-effort data:

Region	Macro F_1	High-risk
Brazil EEZ	0.993	14
Philippine EEZ	0.992	7
Strait of Malacca	0.999	20
Gulf of Guinea	0.992	20

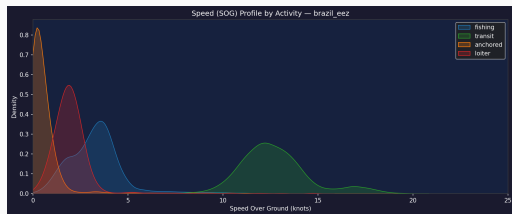
Speed profile KL-divergence vs. GFW real data:
mean $D_{KL} < 0.15$ nats across all gear types.

Gulf of Guinea and Malacca have the highest high-risk counts — consistent with known IUU and STS activity in those corridors.

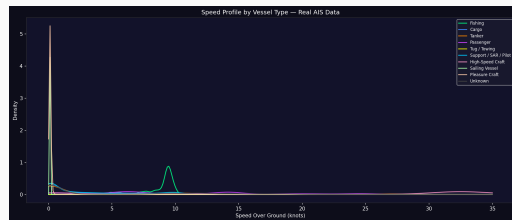


*Activity distribution — Brazil EEZ simulation.
Fishing and transit dominate; loitering and STS
are rarer events.*

Operational Scenarios: Speed Profiles

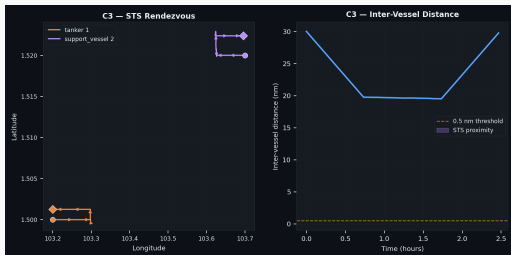


Speed-over-ground profiles by activity class — Brazil EEZ. Fishing (2–4 kn trawling), transit (10–14 kn), anchored (< 0.5 kn).



SOG distributions by vessel type — real NOAA AIS. High-speed craft (HSC) separates cleanly at > 20 kn; pleasure craft and sailing overlap with tug at low speeds.

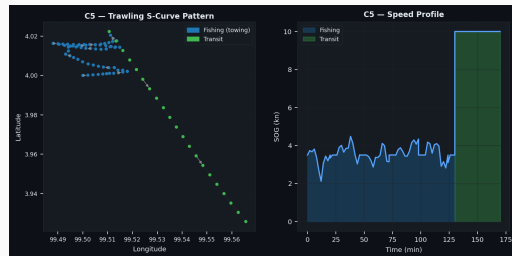
Operational Scenario: STS Transfer Detection



Simulated STS rendezvous — two vessels converging to near-zero relative speed at a common location. The loiter index spikes for both vessels during the transfer window.

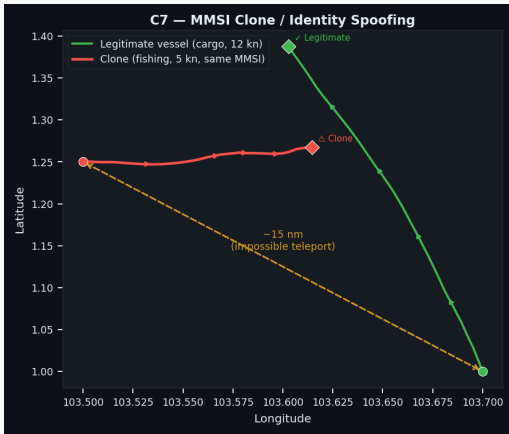
STS scoring:

- Both vessels flagged when loiter index exceeds threshold *and* inter-vessel distance < 500 m
- Composite IUU score elevated by fishing type + dark history

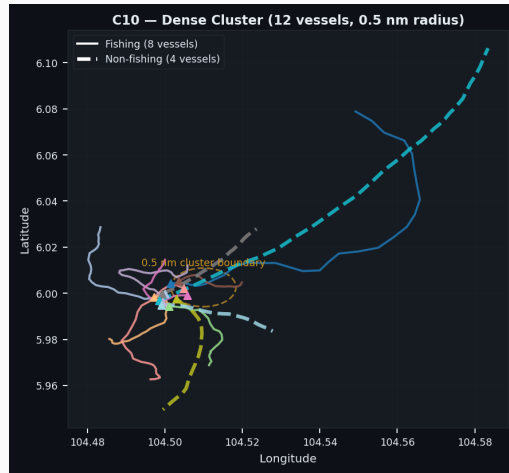


Trawling pattern — characteristic back-and-forth at 2–4 kn with high heading variance. Loiter

Operational Scenario: MMSI Spoofing and Dense Clusters

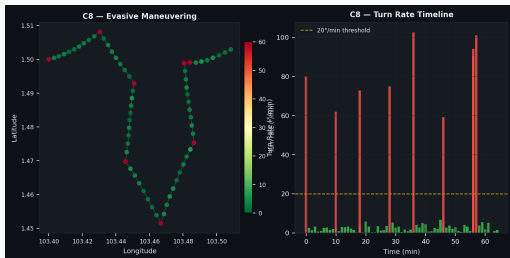


MMSI clone — same identifier appearing at two geographically separated positions simultaneously. Flagged by the dark detector's spoof score.

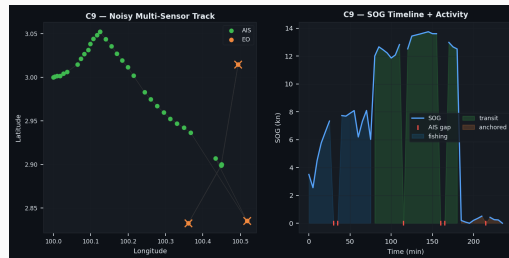


Dense vessel cluster — a common scenario near ports and choke points. Track association is required before per-vessel classification.

Evasive Manoeuvring and Speed Jump Detection



Evasive manoeuvring — irregular course changes with high zig-zag score. Contributes to the spoof component of the dark risk score.



Speed jump with noise — implied speed > 40 kn between consecutive AIS pings. Direct trigger for kinematic impossibility scoring (s_{spoof}).

Discussion: The EO/SAR Generalisation Gap

Open challenge: single EO/SAR observations where `nav_status` and `SOG` are unavailable.

Current behaviour:

- Confidence $c \approx 0.30$ for single EO frame
- Predicted class depends on **spatial context** (proximity to fishing grounds, shipping lanes, time of day)
- Geography prior not yet encoded as a feature

Paths forward:

- SAR foundation models (SARatrX) — pre-trained on unlabelled target data, few-shot vessel recognition
- Maritime geography prior — encode known

Fishing detection in cross-dataset eval:

Dataset	Prec.	Recall
Jun-2023	1.00	≈ 1.00
Dec-2023	1.00	0.12
Jan-2024	1.00	0.18

Zero false positives throughout. Missed cases are fishing vessels **transiting** — the ground truth `nav_status` correctly reads 0 (under way).

The right metric for IUU detection without `nav_status` is **vessel-type** recall ($F_1 = 1.00$), not activity recall.

Limitations and Future Work

Current limitations:

1. **Geography:** real training data confined to US coastal + Eastern Mediterranean. Open-ocean performance extrapolated via simulation.
2. **Temporal:** three daily snapshots — limited seasonal diversity for rare activity classes.
3. **STS/loiter GT:** no real ground truth for STS and loitering in NOAA data; synthetic evaluation only.
4. **Class imbalance:** fishing $< 1\%$ of US coastal AIS traffic. Pacific/Indian Ocean data would improve recall.

Priority future directions:

- Ingest real AIS from Global Fishing Watch API or exactEarth — pipeline is data-agnostic
- SAR/AIS fusion track association layer
- SAR foundation backbone (SARatrX) for EO-only contacts
- Maritime geography prior as feature layer
- Incorporate Pacific/Indian Ocean fishing data to address class imbalance

The feature and classifier pipeline is already designed to accept these data sources — integration is an **engineering** task, not a

Summary of Results

Metric	Value
Activity macro F_1 (real AIS, nav_status available)	1.000
Activity weighted F_1 (cross-dataset)	0.900–0.995
Vessel type macro F_1 (12 classes)	0.969
Synthetic region macro F_1 (4 regions)	0.992–0.999
XGBoost training time (216k examples)	82 s total
Speedup vs. sklearn RandomForest	22×
Inference latency	19 μs/prediction
Dark vessels flagged (Dec 2023 + Jan 2024)	49 vessels
Real vessels across all evaluation datasets	2,820

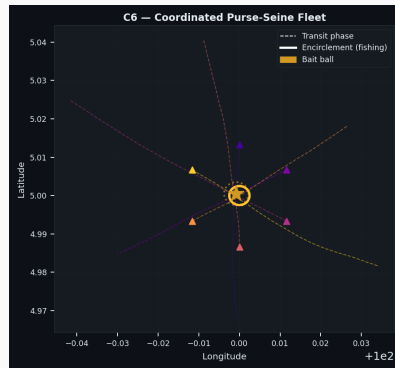
Performance is statistically flat from $N = 1$ to $N = 50$ observations.

No minimum custody duration required for activity classification.

Conclusion

We presented a complete **Maritime Activity Intelligence Engine**:

- 42-feature partial-track vector valid for $N \geq 1$ observations, across AIS, EO, SAR, and radar
- XGBoost with native NaN handling on GPU — no imputation, no minimum track length
- 22 $\times$ training speedup, 19 μ s inference
- Robust across 2,820 real vessels, 3 NOAA datasets, 2 seasons, 2 geographies
- Dark vessel detector with SHAP-interpretable scoring
- Multi-region synthetic validation against GFW fleet data



Coordinated fleet — the target operational scenario the engine was designed to detect and classify.

Code, models, and data loaders available upon request.

Questions?